

Topic: General Fourier Series

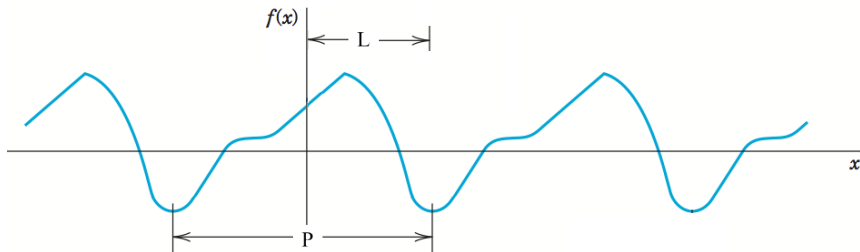
I. Definition

Defn: Suppose that $f(x)$ is periodic with period $P = 2L$. The **Fourier series** of f is

$$f(x) =$$

where d_n are the complex **Fourier coefficients** of $f(x)$ given by

$$d_n =$$



II. Properties of Even and Odd Functions

Type	Definition	Graphical Symmetry	Integral	Example
$g(x)$ is even			$\int_{-L}^L g(x) dx =$	
$g(x)$ is odd			$\int_{-L}^L g(x) dx =$	

Product of odd & even functions			
$f(x)$	$g(x)$	$f(x)g(x)$	<i>Proof:</i> Let $h(x) = f(x)g(x)$
even	even		$h(-x) =$
even	odd		$h(-x) =$
odd	odd		$h(-x) =$

Ex: Identify each of the following functions as either odd, even, or neither

III. Fourier Coefficients for Even and Odd Functions

Rewriting the Fourier Coefficients (d_n) using Euler's Formula ($e^{i\theta} = \cos \theta + i \sin \theta$) we observe

$$\begin{aligned}d_n &= \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi nx/L} dx = \frac{1}{2L} \int_{-L}^L f(x) \left[\cos\left(\frac{\pi nx}{L}\right) - i \sin\left(\frac{\pi nx}{L}\right) \right] dx \\&= \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx + \frac{i}{2L} \int_{-L}^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx\end{aligned}$$

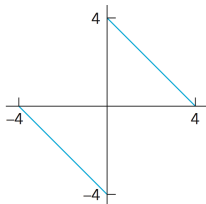
Therefore, if

$$\begin{aligned}\text{if } f(x) \text{ is even: } d_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx = \\d_n &= \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{i\pi nx}{L}\right) dx + \frac{-i}{2L} \int_{-L}^L f(x) \sin\left(\frac{i\pi nx}{L}\right) dx =\end{aligned}$$

$$\begin{aligned}\text{if } f(x) \text{ is odd: } d_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx = \\d_n &= \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{i\pi nx}{L}\right) dx + \frac{-i}{2L} \int_{-L}^L f(x) \sin\left(\frac{i\pi nx}{L}\right) dx =\end{aligned}$$

Summary of Fourier Coefficients when $f(x)$ is odd or even		
$f(x)$ is even	$d_0 = \frac{1}{L} \int_0^L f(x) dx$	$d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{i\pi nx}{L}\right) dx$
$f(x)$ is odd	$d_0 = 0$	$d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{i\pi nx}{L}\right) dx$
$f(x)$ is neither	$d_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$	$d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi nx/L} dx$

Ex: Find the Fourier Series of the given function



· $f(x) =$

· Period =

· $f(x) =$

· $f(x)$ is an function

$$\Rightarrow f(x) = \sum_{n=-\infty}^{\infty} d_n e^{in\pi x/L} \quad \text{where}$$

Ex: Find the Fourier Series of $f(x) = \begin{cases} 0 & -2 \leq x \leq -1 \\ 1 & -1 \leq x \leq 1 \\ 0 & 1 \leq x \leq 2 \end{cases}$ and $f(x) = f(x + 4)$

Notes:

- Since $f(x) = f(x + 4) \implies$ Period =
- $f(x)$ is an function

$$\implies f(x) = \sum_{n=-\infty}^{\infty} d_n e^{in\pi x/L} \quad \text{where}$$

